

HearTMD: Detection of Temporomandibular Disorder Using Hearables

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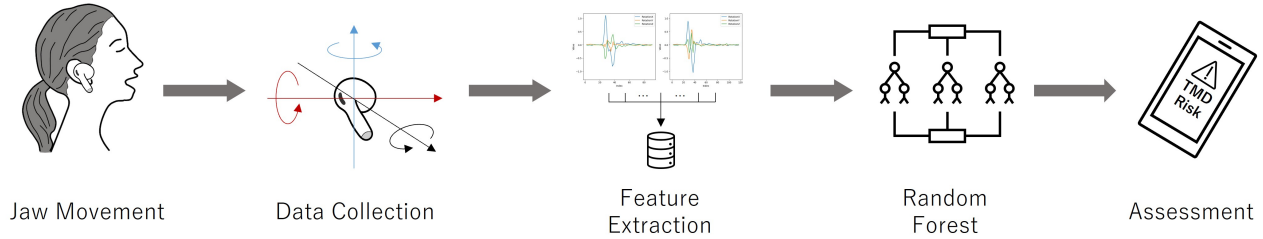


Figure 1: Overview of the proposed system

Abstract

Temporomandibular disorder (TMD) is a chronic disease that presents symptoms such as pain in the temporomandibular joint (TMJ) and masticatory muscles, TMJ noise, and difficulty opening the mouth. It is important to receive appropriate medical examinations and treatment for TMD, as it can interfere with the patient's life if it worsens. In this study, we propose a method to detect TMD from IMU data during jaw movements using hearables. This method enables a non-invasive, low-cost screening in daily life, leading to early detection and health checks. Using data labeled according to the presence or absence of subjective symptoms, the proposed method achieved a sensitivity of 0.641, a specificity of 0.672, and an AUC of 0.700 during maximum mouth opening.

CCS Concepts

• **Human Computer Interaction;**

Keywords

Healthcare, Earables, TMD screening, Non-invasive health monitoring

ACM Reference Format:

Hiyori Tsuji, Takashi Amesaka, Sho Usuda, Wataru Muraoka, Taneaki Nakagawa, and Yuta Sugiura. 2025. HearTMD: Detection of Temporomandibular Disorder Using Hearables. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758416>

1 Introduction

Temporomandibular joint disorder (TMD) is a dental disease that affects the jaw muscles, joints, and connecting fibrous tissues. The main symptoms include pain in the temporomandibular joint (TMJ) and masticatory muscles, TMJ noise, difficulty opening the mouth, and abnormal jaw movement [4]. These symptoms can restrict daily activities and reduce quality of life, making proper diagnosis and treatment essential.

For TMD diagnosis, clinical approaches such as interviews and palpation are combined with imaging techniques like computed tomography (CT) and magnetic resonance imaging (MRI) [9]. However, these methods are limited to medical facilities and involve time and cost for patients. As alternative methods for detecting TMD at a lower cost, thermography camera-based detection [6, 10] and surface electromyography-based detection [7, 8] have been

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UIST Adjunct '25, Busan, Republic of Korea

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ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758416>

Table 1: Detection performance for each dataset

Data	Sensitivity	Specificity	AUC
Grav Acc, Acc, Ang Vel	0.664	0.647	0.682
Grav Acc, Acc	0.827	0.317	0.591
Grav Acc, Ang Vel	0.641	0.672	0.700
Acc, Ang Vel	0.773	0.525	0.680
Grav Acc	0.482	0.572	0.504
Acc	0.782	0.453	0.621
Ang Vel	0.877	0.336	0.648

Grav Acc: Gravitational Acceleration, Acc: Acceleration, Ang Vel: Angular Velocity

proposed. However, their diagnostic efficacy has not yet been fully established. Additionally, these detection methods require specialized equipment, making them difficult to perform easily in any location.

Therefore, this paper proposes a method for detecting TMD using hearables. Hearables are wearable devices worn in the ear and are equipped with multiple sensors, making them useful for measuring human movement and health status [1–3, 5]. Based on the hypothesis that the characteristics of mouth movement differ depending on the presence or absence of TMD, we use hearables capable of measuring the movement of the TMJ and surrounding areas to capture these differences. This method, which can be used in daily life, enables users to perform self-checks easily, contributing to early detection, promoting self-care, and encouraging visits to medical institutions.

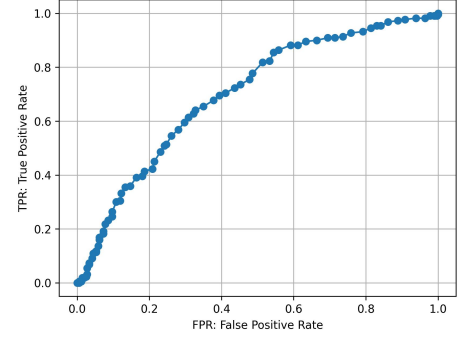
In this paper, we collected data from participants with and without subjective symptoms, aiming to apply the system to actual patients, and constructed a symptom detection model.

2 Implementation

2.1 Data Collection

To build a detection model, data were collected from 11 participants with subjective symptoms (25.1 ± 4.5 years old) and 18 participants without subjective symptoms (22.5 ± 1.8 years old). Subjective symptoms were defined as pain in the TMJ or masticatory muscles, joint noise, difficulty opening the mouth, or abnormal jaw movement. Based on these criteria, we recruited participants who reported such symptoms and collected data accordingly. While the proposed system ultimately aims to detect TMD, expert knowledge suggests that self-recognition of TMD based on subjective symptoms is generally reliable. Therefore, to facilitate efficient data collection, we adopted a strategy in which subjective symptom labels are used for model training, and performance is later evaluated with data from clinically diagnosed patients. In this study, as a preliminary investigation, we assess the detection accuracy of a model that predicts the presence or absence of subjective symptoms. This study was approved by the Ethics Committee of Keio University.

The participants were asked to wear AirPods Pro and perform two types of jaw movements: maximum mouth opening and lateral jaw movements (moving the jaw from side to side), each repeated 20 times. During these tasks, IMU data (angular velocity, acceleration, and gravitational acceleration) were recorded at approximately 25 Hz.


Figure 2: ROC curve for the model using data from maximum mouth opening

2.2 Feature Extraction and Model Construction

In the collected IMU data during jaw movements, time synchronization was first performed. Then, basic statistics (mean, standard deviation, maximum, and minimum) were extracted for each IMU axis and their left-right differences. In addition, for each axis, FFT was applied to extract peak frequency, its amplitude, and the first ten component amplitudes. These features were combined to construct a detection model for determining the presence or absence of symptoms using a random forest classifier. Models were built using different IMU data combinations (single, pairwise, or all three types), resulting in the construction of seven detection models. The combination of angular velocity and gravitational acceleration yielded the highest AUC and was used for the final model (Tab. 1).

A leave-one-participant-out approach was used with the remaining participants split into training and validation sets (five symptomatic and five asymptomatic participants for validation). Five models were trained per fold, and the one achieving the highest validation AUC was applied to the test participant. Sensitivity, specificity, and AUC were computed to evaluate performance, with thresholds determined using the Youden index.

3 Results and Discussion

The model based on maximum mouth opening data demonstrated the highest performance, achieving a sensitivity of 0.641, specificity of 0.672, and an AUC of 0.700 (Fig. 2). In contrast, for lateral movement, the sensitivity was 0.150, the specificity was 0.908, and the AUC was 0.414.

The results indicate that detection is particularly difficult with lateral movement data. According to a specialist in TMJ, even if symptoms are present, it is anatomically unlikely for them to affect lateral movement. In the classification of TMD subtypes, the most common type, myofascial pain disorder (Type I), involves pain and functional limitations in the masticatory muscles rather than in the joint structures. As a result, abnormalities during lateral movements are less likely to be detected. In addition, some participants in the experiment found it difficult to reproduce lateral movement, regardless of the presence or absence of symptoms. Therefore, we believe that the low performance of the model based on lateral movement data can be attributed to these factors.

4 Limitations and Future Work

The detection accuracies of these models remain limited. Therefore, to improve model performance, refinement of useful features is necessary based on further analysis of the data and insights from medical professionals.

In addition, in this study, we constructed models using the presence or absence of subjective symptoms as labels. To verify the effectiveness of these models as screening tools, evaluation with data from actual patients will be necessary in the future.

Furthermore, our future vision includes the possibility of automatic measurement and screening in daily life. The results suggest the potential to detect the presence of TMD during maximum mouth opening. Therefore, the goal is to develop a system with higher usability that can detect based on actions such as eating, talking, and yawning.

5 Conclusion

In this study, we proposed a method to detect TMD using hearables. This approach enables simple and non-invasive health checks by utilizing hearables that are commonly used in everyday life. Future work will focus on improving the detection accuracy of the models and collecting data from clinically diagnosed TMD patients to verify the applicability of the models for screening.

Acknowledgments

Part of this work was supported by KGRI Challenge Grant.

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